

TABLE S3

FEW-SHOT ORIENTED OBJECT DETECTION RESULTS ON ROTsAR-FEW. THE MAIN COMPARISON USES RESNET-50; ADDITIONAL SCL ROWS EVALUATE STRONGER CONVOLUTIONAL AND TRANSFORMER BACKBONES.

				5-Shot			10-Shot			20-Shot		
				bAP	nAP	mAP	bAP	nAP	mAP	bAP	nAP	mAP
Novel Split 1	TFA [10]	ICML 2020	ResNet-50	37.7	6.6	22.2	35.0	8.8	21.9	39.7	14.1	26.9
	BRL [59]	ICASSP 2020	ResNet-50	38.0	6.6	22.3	36.2	8.8	22.5	39.1	14.0	26.6
	TRAML [11]	CVPR 2020	ResNet-50	38.0	5.7	21.9	35.7	8.7	22.2	38.9	13.8	26.4
	DeFRCN [28]	ICCV 2021	ResNet-50	28.0	7.0	17.5	43.3	9.7	26.5	36.7	9.1	22.9
	DCFS [102]	NeurIPS 2022	ResNet-50	14.6	7.9	11.3	26.1	11.7	18.9	19.3	13.4	16.3
	MFDC [30]	ECCV 2022	ResNet-50	42.9	9.5	26.2	48.6	12.1	30.4	43.8	14.5	29.1
	CFA [103]	CVPRW 2022	ResNet-50	37.7	6.5	22.1	36.1	8.7	22.4	39.2	13.3	26.3
	ST-FSOD [25]	TGRS 2023	ResNet-50	48.5	10.5	29.5	51.9	10.6	31.2	45.1	13.1	29.1
	G-FSDet [41]	ISPRS 2023	ResNet-50	42.8	10.4	26.6	39.7	13.6	26.7	44.0	15.3	29.7
	DiGeo [33]	CVPR 2023	ResNet-50	41.4	10.9	26.1	47.3	11.0	29.1	41.2	14.0	27.6
	UNP [73]	TCVST 2024	ResNet-50	37.8	6.6	22.2	36.0	8.8	22.4	39.1	13.7	26.4
	RP-FSOD [43]	TCSVT 2024	ResNet-50	37.9	6.5	22.2	35.2	8.8	22.0	39.0	14.0	26.5
	T-RCNN [104]	TOMM 2024	ResNet-50	32.7	10.8	21.7	37.3	11.6	24.5	39.4	14.2	26.8
	SAE-FSDet [76]	TGRS 2024	ResNet-50	43.7	3.3	23.5	45.9	3.3	24.6	35.5	3.9	19.7
	FOMC [51]	TGRS 2025	ResNet-50	41.4	8.2	24.8	42.5	9.4	26.0	41.7	13.4	27.5
	BDKMR [100]	JAS 2025	ResNet-50	37.7	6.6	22.2	35.4	8.8	22.1	39.0	14.0	26.5
	PoE [74]	TPAMI 2026	ResNet-50	49.4	8.2	28.8	48.3	10.1	29.2	45.2	12.9	29.0
	SCL/ours	Our Method	ResNet-50	60.7	12.6	36.6	60.2	12.4	36.3	58.5	18.0	38.3
	TFA [10]	ICML 2020	ResNet-101	47.6	6.6	27.1	45.4	11.0	28.2	48.0	14.6	31.3
	SCL/ours	Our Method	ResNet-101	60.9	14.5	37.7	60.3	11.0	35.6	59.8	17.2	38.5
Novel Split 2	TFA [10]	ICML 2020	Swin-Large	30.5	10.9	20.7	19.5	13.6	16.6	11.7	14.4	13.1
	SCL/ours	Our Method	Swin-Large	60.0	16.1	38.1	58.4	12.7	35.6	52.2	16.1	34.1
	TFA [10]	ICML 2020	Skysense-Huge	38.0	12.6	25.3	48.4	14.6	31.5	50.1	16.3	33.2
	SCL/ours	Our Method	Skysense-Huge	64.3	<u>14.7</u>	39.5	64.4	17.6	41.0	60.6	20.9	40.7
	TFA [10]	ICML 2020	ResNet-50	49.9	6.1	28.0	48.7	9.9	29.3	47.9	13.9	30.9
	BRL [59]	ICASSP 2020	ResNet-50	49.9	6.1	28.0	48.4	10.0	29.2	47.7	13.8	30.8
	TRAML [11]	CVPR 2020	ResNet-50	50.2	6.0	28.1	48.7	10.0	29.4	48.5	13.6	31.1
	DeFRCN [28]	ICCV 2021	ResNet-50	35.4	8.5	21.9	44.5	9.2	26.9	42.8	10.4	26.6
	DCFS [102]	NeurIPS 2022	ResNet-50	23.7	6.6	15.1	33.5	6.7	20.1	35.9	10.5	23.2
	MFDC [30]	ECCV 2022	ResNet-50	53.4	4.8	29.1	43.9	7.7	25.8	44.4	10.2	27.3
	CFA [103]	CVPRW 2022	ResNet-50	50.1	6.2	28.1	48.7	10.1	29.4	49.8	13.2	31.5
	ST-FSOD [25]	TGRS 2023	ResNet-50	57.6	12.9	<u>35.3</u>	49.8	15.0	32.4	43.4	17.4	30.4
	G-FSDet [41]	ISPRS 2023	ResNet-50	54.6	10.2	32.4	55.0	12.6	33.8	55.1	16.7	35.9
	DiGeo [33]	CVPR 2023	ResNet-50	52.3	10.3	31.3	45.7	12.9	29.3	53.2	16.9	35.0
	UNP [73]	TCVST 2024	ResNet-50	50.0	6.1	28.0	48.7	10.2	29.5	47.8	13.2	30.5
	RP-FSOD [43]	TCSVT 2024	ResNet-50	50.5	6.1	28.3	48.7	10.0	29.4	48.8	13.6	31.2
	T-RCNN [104]	TOMM 2024	ResNet-50	45.1	10.6	27.8	40.0	13.7	26.8	49.7	13.3	31.5
	SAE-FSDet [76]	TGRS 2024	ResNet-50	27.1	6.1	16.6	27.0	9.2	18.1	28.4	9.7	19.1
	FOMC [51]	TGRS 2025	ResNet-50	51.4	8.5	29.9	49.2	12.8	31.0	47.6	15.4	31.5
	BDKMR [100]	JAS 2025	ResNet-50	49.8	6.1	28.0	48.6	9.9	29.3	48.3	13.6	31.0
Novel Split 3	PoE [74]	TPAMI 2026	ResNet-50	50.5	7.6	29.1	50.4	11.2	30.8	49.5	18.8	34.2
	SCL/ours	Our Method	ResNet-50	54.0	16.9	35.5	51.8	<u>16.0</u>	33.9	50.8	<u>24.3</u>	<u>37.5</u>
	TFA [10]	ICML 2020	ResNet-101	50.1	4.3	27.2	49.8	9.5	29.6	47.9	15.1	31.5
	SCL/ours	Our Method	ResNet-101	55.8	<u>13.5</u>	34.7	56.9	13.3	<u>35.1</u>	53.1	20.7	36.9
	TFA [10]	ICML 2020	Swin-Large	41.0	0.3	20.6	38.6	6.2	22.4	39.0	9.2	24.1
	SCL/ours	Our Method	Swin-Large	56.3	8.5	32.4	56.5	17.0	36.8	57.3	25.6	41.4
	TFA [10]	ICML 2020	Skysense-Huge	64.4	1.3	32.8	56.7	3.9	30.3	<u>57.5</u>	8.9	33.2
	SCL/ours	Our Method	Skysense-Huge	<u>62.9</u>	4.7	33.8	64.9	3.1	34.0	63.5	6.7	35.1
	TFA [10]	ICML 2020	ResNet-50	52.5	3.6	28.1	50.0	9.5	29.7	48.1	12.0	30.1
	BRL [59]	ICASSP 2020	ResNet-50	52.7	3.6	28.2	50.6	9.4	30.0	47.9	12.0	30.0
	TRAML [11]	CVPR 2020	ResNet-50	52.7	3.6	28.1	50.6	9.5	30.1	49.9	12.1	31.0
	DeFRCN [28]	ICCV 2021	ResNet-50	37.4	6.4	21.9	49.4	10.8	30.1	51.4	11.4	31.4
	DCFS [102]	NeurIPS 2022	ResNet-50	27.1	6.5	16.8	33.5	8.1	20.8	39.7	9.8	24.7
	MFDC [30]	ECCV 2022	ResNet-50	52.6	6.8	29.7	44.4	7.1	25.8	47.8	9.7	28.8
	CFA [103]	CVPRW 2022	ResNet-50	52.5	3.6	28.1	50.0	9.5	29.7	47.8	12.0	29.9
	ST-FSOD [25]	TGRS 2023	ResNet-50	61.3	7.8	34.6	55.7	11.2	33.5	58.1	15.5	36.8
	G-FSDet [41]	ISPRS 2023	ResNet-50	59.3	6.0	32.6	57.1	10.9	34.0	57.3	14.2	35.8
	DiGeo [33]	CVPR 2023	ResNet-50	54.8	7.1	31.0	47.5	10.7	29.1	57.7	14.0	35.9
	UNP [73]	TCVST 2024	ResNet-50	52.7	3.6	28.2	50.5	9.5	30.0	48.3	12.0	30.2
	RP-FSOD [43]	TCSVT 2024	ResNet-50	52.6	3.6	28.1	50.6	9.4	30.0	47.8	12.0	29.9
	T-RCNN [104]	TOMM 2024	ResNet-50	46.9	8.2	27.5	37.1	11.2	24.2	53.4	8.8	31.1
Novel Split 3	SAE-FSDet [76]	TGRS 2024	ResNet-50	32.2	5.8	19.0	37.8	7.1	22.5	37.5	6.9	22.2
	FOMC [51]	TGRS 2025	ResNet-50	52.2	6.9	29.5	48.6	11.2	29.9	50.7	13.8	32.2
	BDKMR [100]	JAS 2025	ResNet-50	52.8	3.6	28.2	50.3	9.4	29.8	48.1	11.9	30.0
	PoE [74]	TPAMI 2026	ResNet-50	58.5	5.6	32.0	54.1	8.8	31.4	55.8	15.4	35.6
	SCL/ours	Our Method	ResNet-50	62.1	10.9	36.5	56.2	<u>14.6</u>	35.4	53.9	<u>21.2</u>	37.5
	TFA [10]	ICML 2020	ResNet-101	52.5	3.6	28.0	53.1	8.9	31.0	52.8	11.8	32.3
	SCL/ours	Our Method	ResNet-101	65.1	7.6	36.3	<u>61.0</u>	13.8	<u>37.4</u>	59.6	21.6	40.6
	TFA [10]	ICML 2020	Swin-Large	43.3	6.2	24.7	39.0	6.2	22.6	37.8	8.2	23.0
	SCL/ours	Our Method	Swin-Large	<u>64.8</u>	<u>10.6</u>	37.7	59.1	14.9	37.0	59.7	17.1	<u>38.4</u>
	TFA [10]	ICML 2020	Skysense-Huge	68.0	4.0	36.0	57.2	6.4	31.8	<u>62.9</u>	9.9	36.4
	SCL/ours	Our Method	Skysense-Huge	<u>67.4</u>	7.4	<u>37.4</u>	63.8	13.7	38.8	63.0	9.6	36.3

bAP, nAP, and mAP denote AP over base classes, novel classes, and all classes, respectively. **Bold** and underline indicate the best and second-best results within each split.